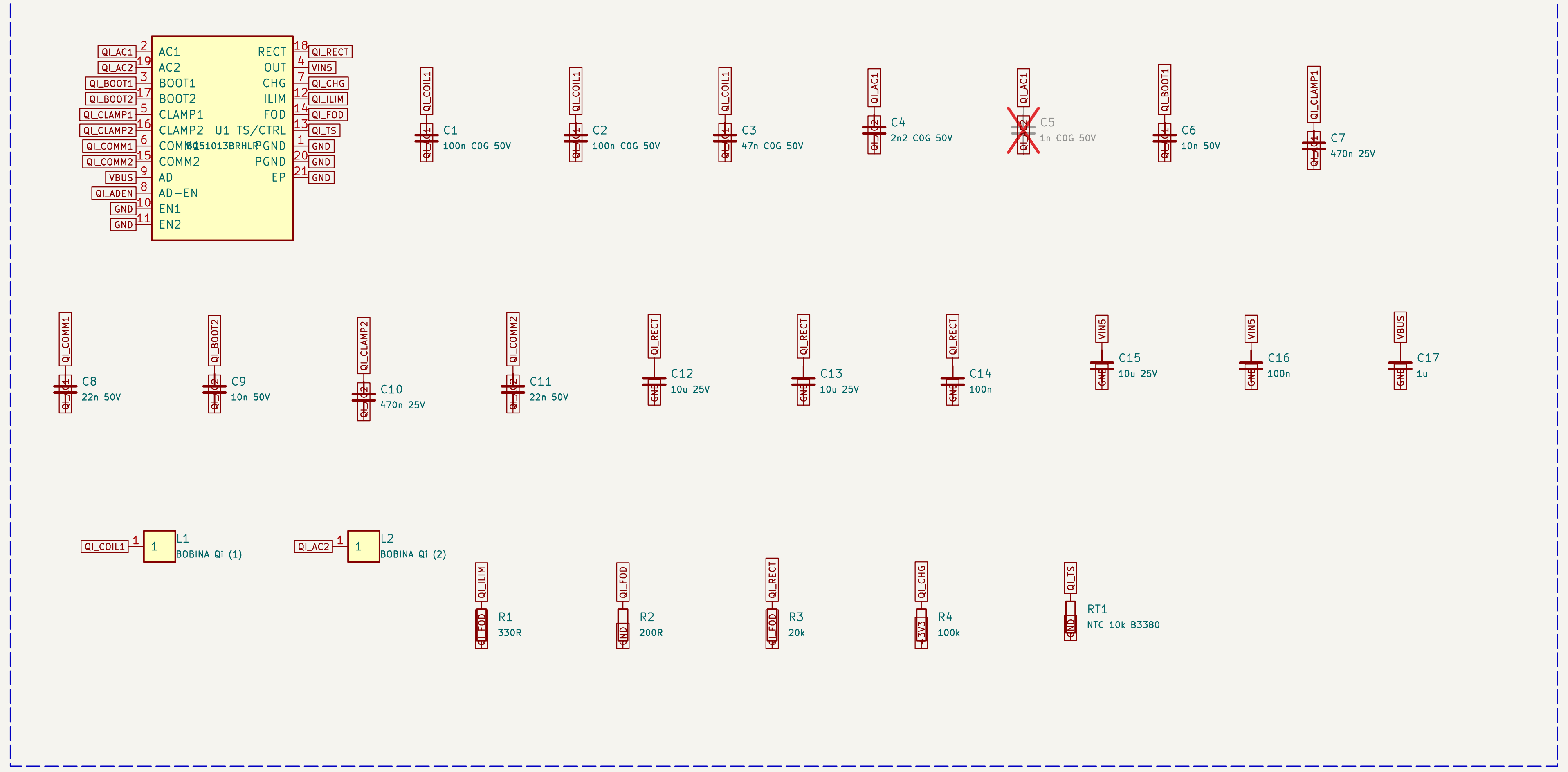
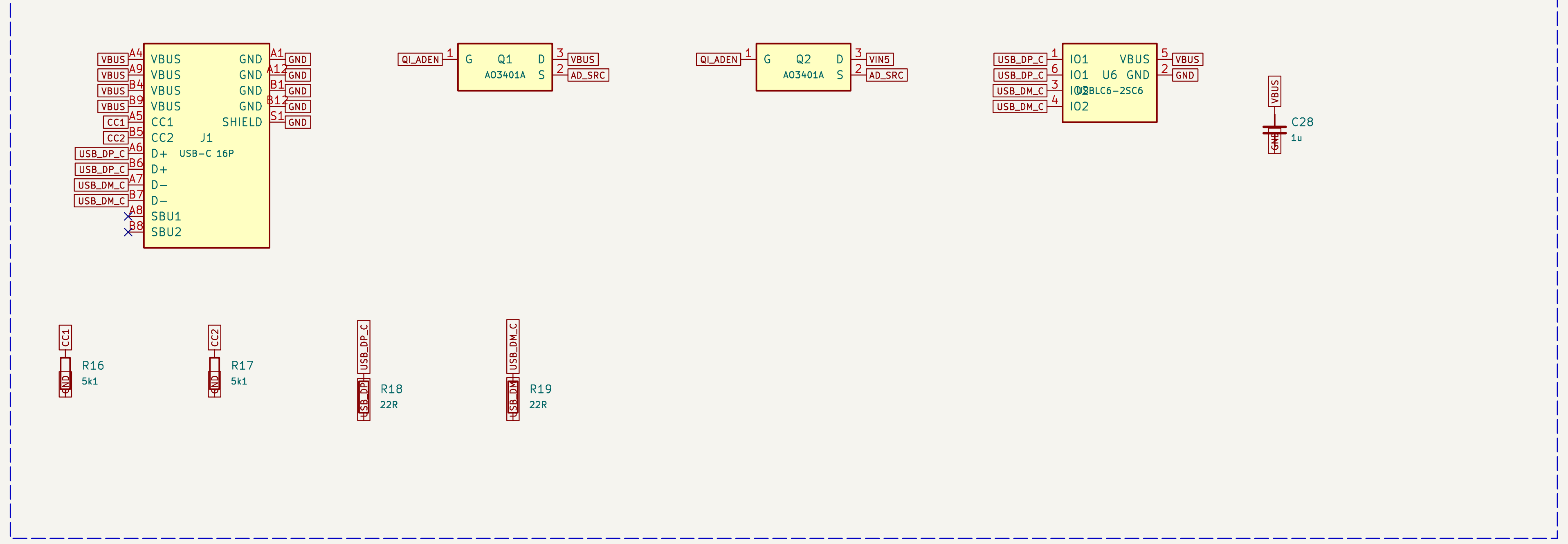
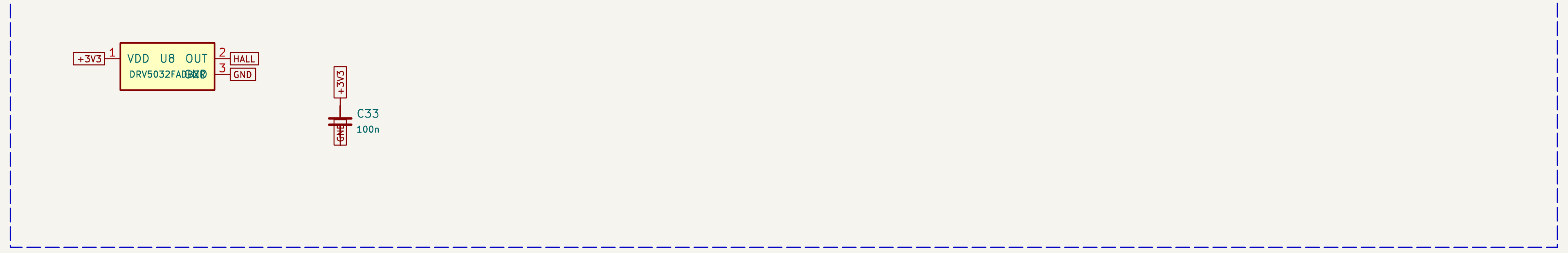
RECEPTOR Qi (BQ51013B) · bobina 30 × 23 bajo la placa · USB por AD/AD-EN


U1 BQ51013BRHLR
AC1 AC2 BOOT1 BOOT2 CLAMP1 CLAMP2 COMM1 COMM2 AD AD-EN EN1 EN2
RECT OUT CHG ILIM FOD TS/CTRL PGND PGND EP
QI_AC1 QI_AC2 QI_BOOT1 QI_BOOT2 QI_CLAMP1 QI_CLAMP2 QI_COMM1 QI_COMM2 VBUS QI_ADEN GND GND
QI_RECT VIN5 QI_CHG QI_ILIM QI_FOD QI_TS GND GND GND
C1 100n C0G 50V
C2 100n C0G 50V
C3 47n C0G 50V
C4 2n2 C0G 50V
C5 1n C0G 50V
C6 10n 50V
C7 470n 25V
C8 22n 50V
C9 10n 50V
C10 470n 25V
C11 22n 50V
C12 10u 25V
C13 10u 25V
C14 100n
C15 10u 25V
C16 100n
C17 1u
L1 BOBINA Qi (1)
L2 BOBINA Qi (2)
R1 330R
R2 200R
R3 20k
R4 100k
RT1 NTC 10k B3380

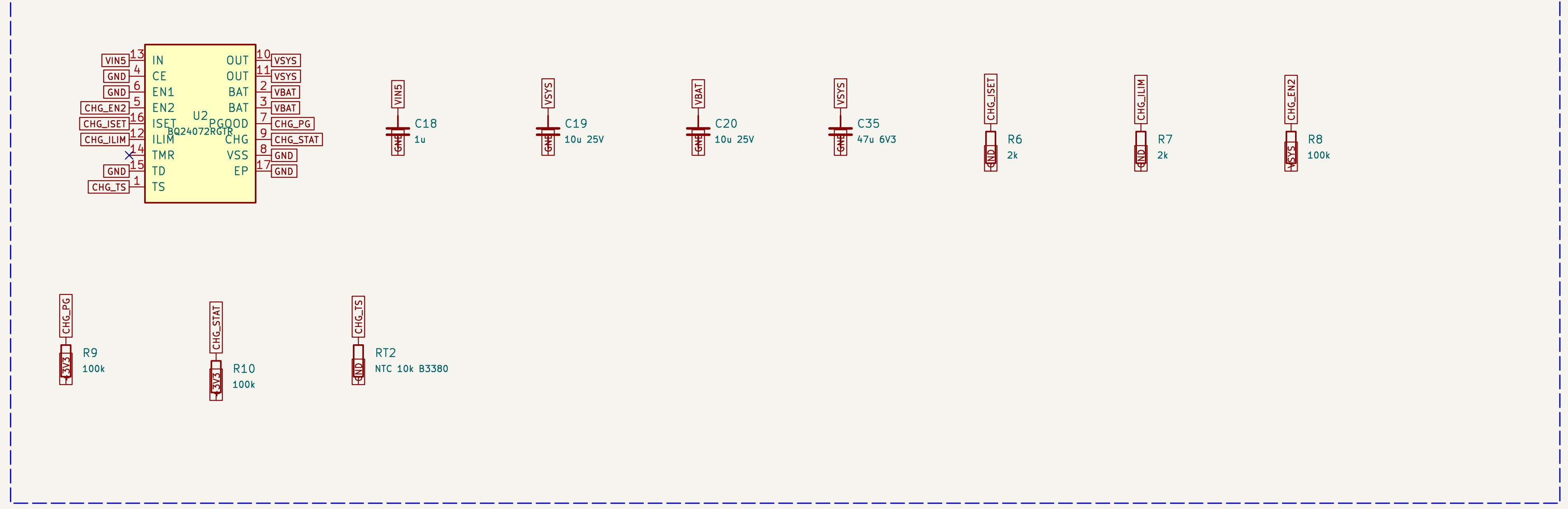
USB-C (programación y carga) · ESD


J1 USB-C 16P
VBUS VBUS VBUS VBUS CC1 CC2 D+ D+ D- D- SBU1 SBU2
GND GND GND GND SHIELD
USB_DP_C USB_DM_C CC1 CC2
Q1 AO3401A
QI_ADEN VBUS AD_SRC
Q2 AO3401A
QI_ADEN VIN5 AD_SRC
U6 USBLC6-2SC6
IO1 IO1 IO2 IO2 VBUS GND
C28 1u
R16 5k1
R17 5k1
R18 22R
R19 22R

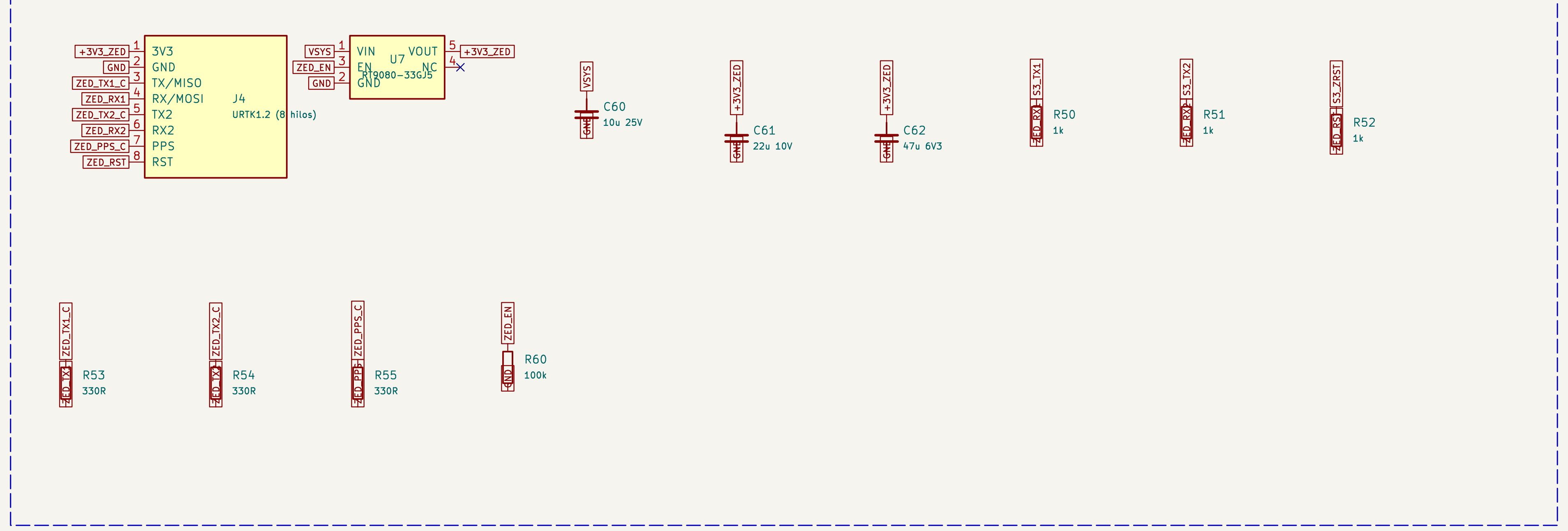
SENSOR DE SOPORTE: Hall DRV5032FA (imanes del soporte)


U8 DRV5032FADBZR
VDD OUT GND
+3V3 HALL GND
C33 100n

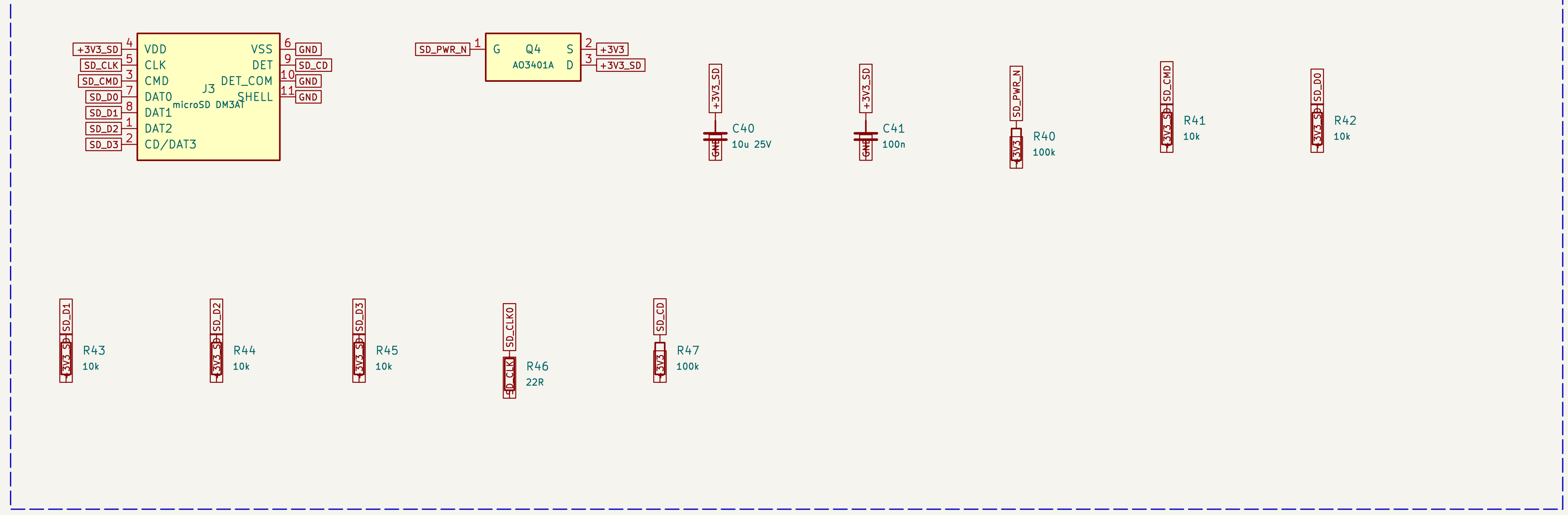
CARGADOR CON CAMINO DE POTENCIA (BQ24072) · batería 1S 2000 mAh (654060)


U2 BQ24072RGTR
IN CE EN1 EN2 ISET ILIM TMR TD TS
OUT OUT BAT BAT PGOOD CHG VSS EP
VIN5 GND GND CHG_EN2 CHG_ISET CHG_ILIM GND CHG_TS
VSYS VSYS VBAT VBAT CHG_PG CHG_STAT GND GND
C18 1u
C19 10u 25V
C20 10u 25V
C35 47u 6V3
R6 2k
R7 2k
R8 100k
R9 100k
R10 100k
RT2 NTC 10k B3380

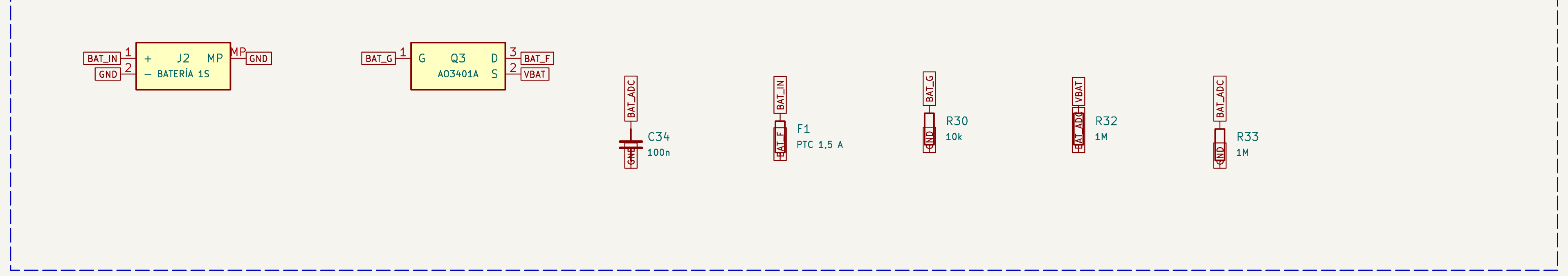
URTK1.2 / ZED-F9P: 3,3 V propio con EN (RT9080) · UART1 · UART2 · PPS · RESET


J4 URTK1.2 (8 hilos)
3V3 GND TX/MISO RX/MOSI TX2 RX2 PPS RST
+3V3_ZED GND ZED_TX1_C ZED_RX1 ZED_TX2_C ZED_RX2 ZED_PPS_C ZED_RST
U7 RT9080-33GJ5
VIN EN GND VOUT NC
VSYS ZED_EN GND +3V3_ZED
C60 10u 25V
C61 22u 10V
C62 47u 6V3
R50 1k
R51 1k
R52 1k
R53 330R
R54 330R
R55 330R
R60 100k

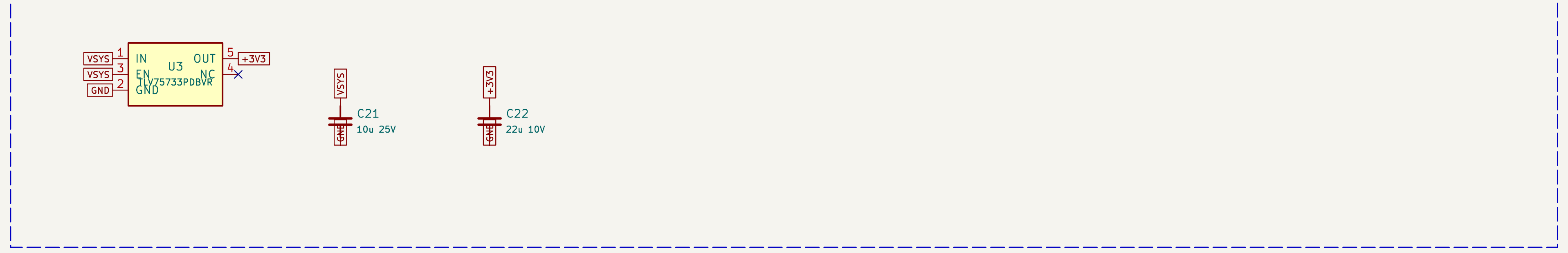
microSD (SDMMC 4 bits) · alimentación conmutada


J3 microSD DM3AT
VDD CLK CMD DAT0 DAT1 DAT2 CD/DAT3
VSS DET DET_COM SHELL
+3V3_SD SD_CLK SD_CMD SD_D0 SD_D1 SD_D2 SD_D3
GND SD_CD GND GND
Q4 AO3401A
SD_PWR_N +3V3 +3V3_SD
C40 10u 25V
C41 100n
R40 100k
R41 10k
R42 10k
R43 10k
R44 10k
R45 10k
R46 22R
R47 100k

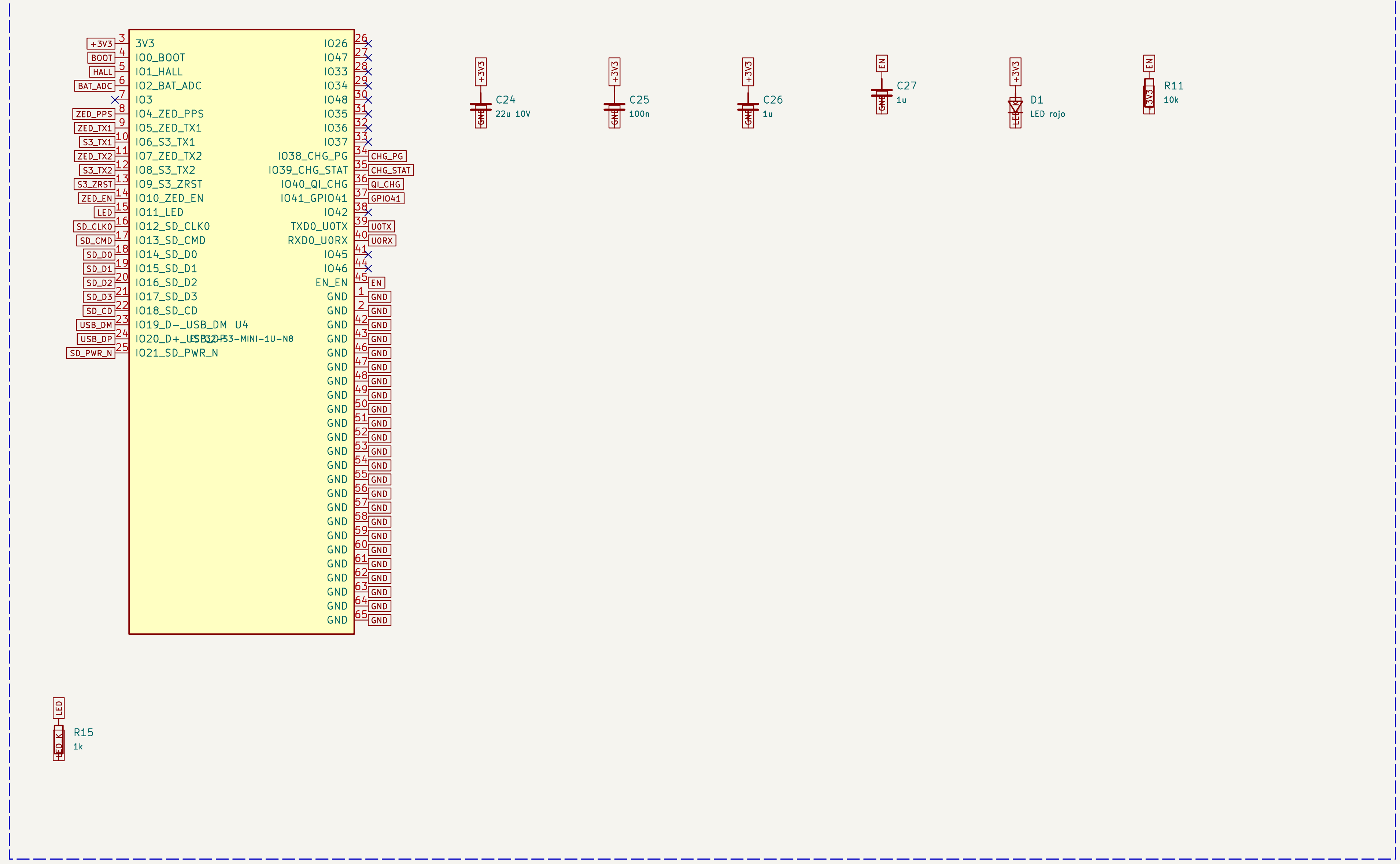
BATERÍA: PTC + protección de polaridad · medida por ADC


J2 BATERÍA 1S
+ − MP
BAT_IN GND GND
Q3 AO3401A
BAT_G BAT_F VBAT
C34 100n
F1 PTC 1,5 A
R30 10k
R32 1M
R33 1M

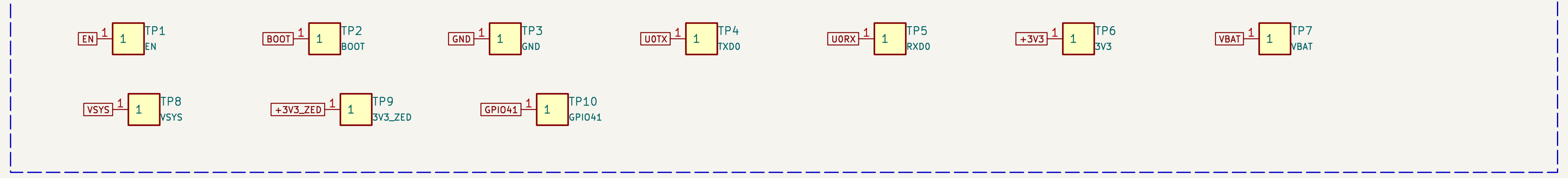
3,3 V PRINCIPAL (TLV75733P, 1 A)


U3 TLV75733PDBVR
IN EN GND OUT NC
VSYS VSYS GND +3V3
C21 10u 25V
C22 22u 10V

ESP32-S3-MINI-1U-N8 · arranque · LED


U4 ESP32-S3-MINI-1U-N8
3V3 IO0_BOOT IO1_HALL IO2_BAT_ADC IO3 IO4_ZED_PPS IO5_ZED_TX1 IO6_S3_TX1 IO7_ZED_TX2 IO8_S3_TX2 IO9_S3_ZRST IO10_ZED_EN IO11_LED IO12_SD_CLK0 IO13_SD_CMD IO14_SD_D0 IO15_SD_D1 IO16_SD_D2 IO17_SD_D3 IO18_SD_CD IO19_D-_USB_DM IO20_D+_USB_DP IO21_SD_PWR_N
IO26 IO47 IO33 IO34 IO48 IO35 IO36 IO37 IO38_CHG_PG IO39_CHG_STAT IO40_QI_CHG IO41_GPIO41 IO42 TXD0_U0TX RXD0_U0RX IO45 IO46 EN_EN GND
+3V3 BOOT HALL BAT_ADC ZED_PPS ZED_TX1 S3_TX1 ZED_TX2 S3_TX2 S3_ZRST ZED_EN LED SD_CLK0 SD_CMD SD_D0 SD_D1 SD_D2 SD_D3 SD_CD USB_DM USB_DP SD_PWR_N
CHG_PG CHG_STAT QI_CHG GPIO41 U0TX U0RX EN GND
C24 22u 10V
C25 100n
C26 1u
C27 1u
D1 LED rojo
R11 10k
R15 1k

PUNTOS DE PRUEBA


TP1 EN
TP2 BOOT
TP3 GND
TP4 TXD0
TP5 RXD0
TP6 3V3
TP7 VBAT
TP8 VSYS
TP9 3V3_ZED
TP10 GPIO41
EN BOOT GND U0TX U0RX +3V3 VBAT VSYS +3V3_ZED GPIO41

Generado por fuente/avo_back_v4_pcb.py (mismo origen de datos que la placa). Las conexiones son etiquetas globales: dos pines con la misma etiqueta están unidos.

AVO Performance Horse
Sheet: /
File: AVO_BACK_V4_PCB.kicad_sch
Title: AVO BACK V4 · placa de la espalda · REV A
Size: User Date: 2026-10-29 **Rev: A**
KiCad E.D.A. 9.0.9 Id: 1/1